

Abstract

This paper reviews the literature regarding the effect of taking additional math in high school on college success. Researchers have established that taking additional math classes in high school is strongly correlated with success in college and higher earnings. More recent research seeks to show that taking additional math in high school actually *causes* success in college. Such analyses are difficult because researchers must draw meaning from naturally occurring data, rather than through experimentation. Researchers have employed a few different methods of causal inference with varying levels of success. Studies using the best methods suggest that taking additional math in high school mathematics does in fact cause an increase in college enrollment and wages, but more studies should be done to confirm these findings. The paper ends by discussing areas for further research.

Keywords: curriculum effects, mathematics education, education policy, education reform, high school mathematics, algebra, geometry, college success indicators, economics of education, causal inference

The Effect of Additional Math in High School on College Success

Education is paramount to the success of society. Individuals in a more educated society can understand and take care of each other. An educated individual provides for their own needs and leads a more fulfilling life. This literature review explores the effect taking additional math classes in high school has on college success. Knowing if mathematics has a positive impact on overall college and career success is important because if it does, both government and privately funded educational institutions should consider implementing policies that raise math teacher salaries, improve teacher training, and fund more studies about how to teach math well.

For years, mathematics has been one of the core subjects taught in school. Success in Mathematics is often viewed as being critical to overall success in school. Does taking additional math classes in high school cause college success? Many studies have shown a strong correlation between additional math course taking and overall educational success (Altonji, 1992; Gamoran & Hannigan, 2000; Levine & Zimmerman, 1995; Smith 1996). However, these results do not imply that additional mathematics causes overall educational success. This question is difficult to answer, because a study must differentiate somehow between students whose abilities make them more likely to both succeed in college and take more math, and students who become more prepared for college by taking more math. To account for these confounding variables, some researchers use methods of causal inference such as propensity score matching, instrumental variables, and regression discontinuity designs.

This paper takes a deeper look into why it is so difficult to discover if additional high school mathematics causes college success and examines the methods that have been used to overcome these difficulties. Certainly, taking additional math courses in high school is correlated with college success. Studies using the most cutting-edge research methods agree that taking

additional math in high school increases college enrollment and wages. Some studies also find that additional math increases college completion. Researchers should focus on utilizing sound statistical methods in future studies to confirm these findings.

Methods and Related Work

The literature reviewed herein was found through keyword searches in Google Scholar and EBSCO Academic Search Premier using phrases such as “math success effects,” “math curriculum effects,” “effect of high school math success on later educational success,” and similar phrases. Prior to using keyword searches, the author was not familiar with any literature on this topic. Literature was reviewed from any discipline, as long as it was relevant to the topic at hand. Additional papers were found and reviewed in the references of papers found by keyword searches. In total, about 45 papers were reviewed, with 8 being selected for inclusion. The 8 papers included in this review are, to the authors knowledge, the only papers in the literature where the authors estimated the *causal* effect of additional math courses in high school on college success using some method of causal inference. The key details of each of the included papers, including the discipline(s) of the authors, are shown in Table 1. There is a substantial body of research that addresses the relationship between high school curriculum and college outcomes without estimating causation (only looking at correlation). This work has been excluded, as it is not the focus of the present review. Those wishing to explore this body of research could start by reading Adelman (2006), Aughinbaugh (2012), and Stein, Kaufman, Sherman, and Hillen (2011) and following the references provided therein.

Most of the literature that seeks to establish the causal relationship between additional math classes in high school and college success was published in journals and written by authors working in the fields of education and the economics of education. Many of the papers on this

topic have brief literature reviews in their introduction, but none of them have the same scope or depth as the present review. Stein et al. (2011) conducted a literature review specifically to understand more about the universal algebra movement: who is taking algebra by the eighth or ninth grade, and what are the outcomes for these students? They conclude that taking algebra by the eighth or ninth grade is strongly correlated with later high school and college success, but only for students who are prepared for it, or who are given the extra support that they need to succeed. Their study differs from the present study in that their sole focus is on early access to algebra, and they include studies whether or not they made an effort to show causation rather than just correlation.

The Inherent Problems

Even Altonji (1992), who largely pioneered research in the effects of curriculum design on college and career success, recognized the multitude of problems that come with seeking to estimate the effect of additional high school mathematics on college success. All of the literature that seeks to show that additional high school math causes college success touch on some of these difficulties (Altonji, 1992; Attewel & Domina, 2008; Byun, Irvin, & Bell 2015; Dougherty, Goodman, Hill, Litke, & Page, 2017; Joensen & Neilsen, 2009; Kim, Kim, DesJardins, and McCall 2015; Levine & Zimmerman, 1995; Rose & Betts, 2001). However, the most thorough expositions of these issues are given by Altonji (1992) and Allensworth, Nomi, Montgomery, & Lee (2009). They point out selection bias on an individual level, selection bias on a school level, effects of socioeconomic standing, effects of parents, and differences in instructor quality.

As an example of selection bias in this context, consider that Kim et al. (2015) showed that students who pass Algebra II in high school are more likely to attend college. Without further analysis, we cannot use this fact to argue that the math class caused them to attend

college because the sample of students who took Algebra II is not random: more advanced students are naturally more likely to take more advanced classes. Furthermore, even using students' prior test scores to control for the lack of random selection can be dangerous, as many skills that help students succeed go unmeasured, especially before high school (Altonji, 1992). As Altonji (1992) noted, parents who push their children to attend college are also likely to pressure their children to take higher levels of math in high school. This student level selection bias plagues the current study.

Byun et al. (2015) declare that assuming advanced math courses have the same benefits for students of different socioeconomic backgrounds is dangerous. Thus, researchers should control for student backgrounds in their analyses. Though disregarding students' socioeconomic backgrounds will still give information about the effects of the courses, researchers will gain more information and more accurate information by considering it as part of the analysis.

Now, suppose that to avoid student level selection bias researchers compare students across schools. For example, they could compare college success for students who attended a school that requires Algebra II to college success for students who attended a school that does not require Algebra II. Comparing across schools still fails to solve the problem, because the schools are not randomly selected. Allensworth et al. (2009) pointed out that most likely, the school that requires Algebra II also has a stronger culture of preparing students for college. This method introduces a school level selection bias.

Altonji (1992) brings up the point that differences in difficulty or quality of instruction in a course can have unseen effects on a student's later success. These disparities create bias across schools and even across teachers at the same school. Additionally, standards for what a student must understand to pass a given class vary across states, schools, teachers, and, as Schneider

(2009) pointed out, across years. Thus, it is difficult to measure the effects of taking additional math courses because it is difficult to measure what a student has actually learned in comparison to another student. Kim et al. (2015) control for the issue by studying subjects who can be compared to one another by some standardized test which all students took. In this case researchers must recognize that even standardized tests give an imperfect representation of student knowledge.

Methods Used in Overcoming the Problem

Clearly, many factors confound the data about the effects of mathematics education on college success. There are enough confounding factors that enumerating them all is impossible. Thus, it is critical to use methods that will control for the seen and unseen factors affecting the data. The most robust way to establish causation between variables is through randomized trial (Angrist & Pischke, 2015, Mosteller & Boruch, 2002). To deduce the causal effect of taking additional math classes in high school on college success, a randomized trial would mean constructing an experimental design that involves randomly assigning students to take additional or more advanced math classes. The political barriers to constructing such an experiment are very high, and some would argue that doing so is unethical (Mosteller & Boruch, 2002). Likely for these reasons, such an experiment has not been conducted. Instead, researchers must use natural experiments: naturally occurring situations where students were randomly or pseudo-randomly assigned to taking an additional or a higher-level math class (DiNardo, 2018). Methods used for analyzing data in such a way as to estimate a causal effect are known as methods of causal inference. Such methods are practiced by researchers in many fields including statistics, economics, and epidemiology. The reader can refer to Pearl (2009) for an introduction to methods of causal inference generally, or Angrist & Pischke (2015) for an overview of uses of

causal inference in the social sciences in particular. Here we review which methods of causal inference have been applied to show if taking additional math classes in high school causes college success.

Matching

Matching techniques seek to obtain a sense of random assignment by matching each experimental subject with another experimental subject of similar background who was not given treatment (Todd, 2018). One commonly used type of matching is Propensity Score Matching (PSM), where subjects are matched using a propensity score: the probability that they would receive treatment based on preexisting factors (Hahn, 2018). For example, suppose researchers want to study the effects of taking Algebra II in high school. Using PSM, to see the effect of having taken Algebra II, they would match each student who took Algebra II with a student who did not take Algebra II of similar socioeconomic standing, parental attitudes, past performance, and all other possible characteristics that could affect the study. One of the drawbacks of matching techniques is that they only control for measurable variables, so they do not solve the fundamental problem of selection bias: that subjects make selections based on unobservable characteristics (Todd, 2018).

Attewell and Domina (2008) used propensity score matching to analyze data from a longitudinal study done by the U.S. Department of Education starting with eighth graders in 1988. They used parental education, income, employment status, and marital status along with student educational expectations and prior academic performance to create their propensity scores, then compared students with similar scores who took a slightly different curriculum in high school than the other, with one including higher level math. They found that students who took a more

intense curriculum in High School math and English were more likely to attend college, but were not necessarily more likely to graduate.

More recently, Byun et al. (2015) utilized PSM to analyze data from the Educational Longitudinal Study of 2002-2006. They matched students on a wide array of characteristics, including socioeconomic status, prior academic achievement, and parental educational expectation. They found the positive impact that taking advanced math classes has on college enrollment can mostly be explained by other factors such as family and school influences. They also found that taking advanced math classes has less of a positive impact for Black students than for White students. However, as previously mentioned, since PSM relies on matching based on observable characteristics, there is always some level of inaccuracy due to unobserved characteristics. As we will see, there are other methods that control for both measured and unmeasured confounding factors.

Instrumental Variables

Instrumental variable estimation is another way of estimating a causal relationship between two variables when a randomized trial is not feasible. In order for an instrumental variable estimation to establish causation, there must be some variable, called an instrumental variable, which meets two conditions: (1) it is correlated with the explanatory variable and (2) it is not correlated with the outcome variable except by virtue of its association with the explanatory variable (see Woolridge, 2013, for a proof that a variable fitting these requirements can be used to estimate causation). In estimating the effect of additional math in high school on college success, the explanatory variable is the math classes taken by a student in high school and the outcome variable is how well the student performs in college. Therefore, a valid instrument must

be correlated with high school math curriculum but not with college success (except through its correlation with high school math courses).

Altonji (1992) introduced the idea of using variation of curriculum across high schools (specifically, number of courses taken in each subject) as an instrumental variable. By analyzing data from the National Longitudinal study of 1972, he found that taking one additional academic course in high school yields little effect on future success, but the effect of taking additional math is greater than the effect of taking more of any other subject. His methods rely on two assumptions: that schools offer identical courses to students but require different courses than one another and that the differences in course requirements are unrelated to student characteristics and any unobserved characteristics (Altonji 1992). By his own admission, these assumptions are not realistic. This highlights one of the primary weaknesses of the instrumental variables approach: the causation argument only holds true if the instrumental variable is correlated with the explanatory but otherwise not correlated with the outcome variable, and such an instrumental variable is often difficult to find.

Levine and Zimmerman (1995) used similar methods to Altonji, with the same instrumental variable, to analyze data from the National Longitudinal Survey of Youth (NLSY) and High School and Beyond (HS&B). In the NLSY, they found no effect for women but found that men who take more math classes are more likely to attend college, but are no more likely to graduate. In the HS&B data, they found no effect for men, but that women who take more math in high school are more likely to attend and graduate from college. Across both data sets, they found that additional math in high school increased the wages for women who graduated from college.

Rose and Betts (2001) also use data from HS&B and similar instrumental variable methods as Altonji. They also improve on Altonji's method by adding more controls for school effects, helping the instrumental variables assumptions that he uses become more believable. They showed that taking higher level math classes in high school has a significant effect on college success. According to their results, taking higher level math in high school had a stronger positive effect on college success for women.

More recently, Joensen and Nielsen (2009) have used instrumental variables techniques to estimate the effect of additional math on college success. However, the natural experiment that they took advantage of doesn't require such strong assumptions as Altonji's. They studied high school students in Denmark. In the 1980s, there was a pilot program for a new curriculum instituted at some schools—a curriculum giving students more flexibility in their course options. In the pilot program, some students took more advanced math than they would have without it. This natural experiment does have some bias, as the schools self-selected to be a part of the pilot program and students still selected to take math. As their instrumental variable, Joensen and Neilson (2009) use whether or not a student's high school decided to enroll in a pilot program *after* the student was enrolled in the high school. They add additional controls for the preexisting qualifications of students. They find that those who take more mathematics than they would have without the pilot curriculum received a higher education and have higher earnings later on.

Kim et al. (2015) analyzed data regarding students in Florida, following them from grades 7-12 and into college and the workplace. They used county unemployment levels during the ninth grade year of the subjects of interest as their instrumental variable. They concluded that completing Algebra II in high school has a significant impact on a student deciding to attend college, but no impact on their completion of college. However, it is debatable if their

instrumental variable meets the criteria for being a valid instrument. It seems unlikely that unemployment during a student's ninth grade year has no correlation with their decision to attend and perform well in college. They support their claim by showing that the unemployment rate changes enough from year to year that it won't have an impact. However, if unemployment has an impact on what math classes a student chooses to take in ninth grade, as they claim it does, it would be likely that it also impacts many other variables that would affect student's decision to attend and perform well in college.

Regression Discontinuity Design

Using a regression discontinuity design (RD design) is another way of extracting randomness from a situation that does not seem random. Regression discontinuity design is achieved by comparing only subjects close to some arbitrary cutoff to obtain a sense of randomness (Lee & Lemieux, 2010). Consider that students are often assigned to take certain courses based on a placement test. Assigning students to take a different class based on a test is not random assignment. However, it could be considered random assignment for those students who scored close to the cutoff on the test. Suppose that students who scored 80% or higher on the placement test were placed into the higher-level math class. On average, the students who scored 79% and the students who scored 80% have nearly identical mathematical abilities. Thus, by only comparing students who scored close to the cutoff, researchers can obtain a sense of random assignment—this is the strength of using an RD design (Lee & Lemieux, 2010). In order for a regression discontinuity design to estimate causation, the subjects must not be able to precisely control the variable that decides which subjects receive treatment, and all other factors related to the outcome variable must be continuous with respect to the treatment variable. Compared to the assumptions necessary to establish causation under other methods of causal

inference, these assumptions are much easier to satisfy, and much easier to test (Lee & Lemieux, 2010). Therefore, causal effects that are estimated using an RD design are often more robust than those using other methods.

Dougherty et al. (2017) observed the results of a pilot program in the Wake County Public School System (North Carolina) that placed many students on a higher math track than usual based on a placement test. They found that taking a more rigorous math curriculum in middle school causes students to obtain higher composite scores on a college preparedness that tested math, science English, and reading, with not all of the increase in their score coming from the math portion. The student's intent to attend college also increased.

Because of the great strength of the RD design in estimating causal effects with few assumptions, more researchers should employ an RD design in examining the effect of high school math curriculum on college success. The existing study by Dougherty et al. (2017) uses sound methods to estimate causation, but it unfortunately only measures intent and preparedness to attend college, rather than examining actual college enrollment and completion data. A study that uses similar methods to estimate causation but follows the students all the way through college enrollment and completion would be a great addition to this body of work.

Discussion

The aggregated results of the surveyed literature are displayed in Table 1. As shown, there is a consensus that taking additional math in high school increases college enrollment and wages, but its effect on the completion of college is disputed. A few of the studies draw attention to a difference in curriculum effects for students of different gender, race, or socioeconomic status (Byun et al., 2015; Dougherty et al., 2017; Levine & Zimmerman, 1995; Rose & Betts, 2001). However, there is not enough consistency across studies to say with certainty which

particular group receives less of a benefit from treatment. Future research should seek to better understand under what circumstances a more rigorous high school math curriculum is not creating value for some students, so that future effort can focus on remedying this situation.

In his book, *The Math Myth*, Andrew Hacker (2016), a Political Scientist, argues that the only reason success in mathematics correlates so strongly with college success is that the current educational system mandates math for college entrance exams and general education requirements. Although he does not have evidence to support his claim, he does pose an interesting question that is worth investigating: Would success in high school mathematics lead to college success even if it was not used by colleges to decide who to admit? Attewell and Domina (2008) argue that since a more rigorous curriculum improved college success for all students, not just those already in higher curriculum brackets, the effects are due to an actual improvement of the students' abilities, and not just because colleges look to accept students with certain math credentials. However, this argument seems tenuous and should be backed by further study.

Conclusion

It is important to understand the impact that mathematics education has on overall educational success so that local, state, and national leaders can make better policy and curriculum decisions. The strong correlation between taking additional math in high school and college success has been known for some time. More recent studies done with various methods of causal inference suggest that taking additional math in high school actually causes success in college. Researchers should do more studies to confirm these findings, using natural experiments where a sense of random assignment can be used to establish causation. Further research should also be done on related questions: How can we ensure that all students are reaping the benefits of

a more rigorous high school math curriculum? Would a strong high school math background increase college success even without requiring it on entrance examinations? These questions and their answers should drive future research and policy decisions.

<i>Authors</i>	Discipline of Authors	Dataset(s)	Methods	Treatment	Increases College Enrollment	Increases College Completion	Increases Wages	Different Effects for Different Groups
<i>Altjoni (1992)</i>	Economics	NLS 1972	Instrumental Variables	Additional year of high school math	Yes	Yes	Yes	N/A
<i>Attewell & Domina (2008)</i>	Sociology	NELS 1988	Propensity Score Matching	Taking more advanced curriculum (across 5 levels)	Yes	No	N/A	N/A
<i>Byun, Irvin, & Bell (2015)</i>	Education	ELS 2002	Propensity Score Matching	Completing one class beyond Algebra II in high school	Yes	N/A	N/A	Yes
<i>Dougherty, Goodman, Hill, Litke, & Page (2017)</i>	Economics/ Education	WCPSS longitudinal student information	Regression Discontinuity Design	Placement on a college-preparatory math track in middle school	Yes*	N/A	N/A	Yes
<i>Joensen & Nielsen (2009)</i>	Economics	Danish pilot program 1984-87	Instrumental Variables	Additional high-level math class in high school	Yes	Yes	Yes	N/A
<i>Kim, Kim, DesJardins, & McCall (2015)</i>	Economics/ Education	Florida's student unit record 1994	Instrumental Variables	Completing Algebra II in high school	Yes	No	N/A	N/A
<i>Levine & Zimmerman (1995)</i>	Economics	NLSY 1979, HS&B 1980	Instrumental Variables	Additional year of high school math	Yes	Yes	Yes	Yes
<i>Rose & Betts (2001)</i>	Economics/ Education	HS&B 1980	Instrumental Variables	Additional year of high school math	Yes	Yes	Yes	Yes

*Intent and preparedness to attend college

TABLE 1: Summary of effects of taking additional math in high school

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